

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2019- 2020 (Odd Semester)

Innovative Practices Description

UNIT I DIGITAL FUNDAMENTALS

Degree, Semester & Branch: III Semester B.E. ECE A

Course Code & Title: EC8392 Digital Electronics

Name of the Faculty member: Mrs.S.Jeeva

Name of the Topic: Introduction to Digital World

Name of the Innovative Practice: Multimedia Approach

Date & Duration: 01.07.2019 & 10 minutes

ICT Tool Used: LCD Projector

Description:

Multimedia approach is an *approach* of teaching in which different mediums are incorporated to make the teaching-learning more effective, enthusiastic, etc.

Multimedia, is the combination of various digital media types such as text, images, audio and video, into an integrated multi-sensory interactive application or presentation to convey information to students.

Goals (Learning Outcomes):

- ✓ The students will be able to apply the digital fundamentals.

Use of Appropriate Method:

Justification for choosing the Multimedia approach activity for this topic:

Use of Multimedia (audio-visual aids) help in maintaining discipline in the class since all the students' attention are focused in learning. This interactive session also develops critical thinking and reasoning that are important components of the teaching-learning process. Students learn when they are motivated and curious about something. Traditional verbal instructions can be boring and painful for students. However, use of audio-visual provides intrinsic motivation to students by peaking their curiosity and stimulating their interests in the subjects.

Effective Presentation:

Implementation (Plan & Execution) with Proof:

First day class to create effective learning process by to play multimedia about “How computer works?” from khan academy. It helps to create curiosity in learning process. The students are motivated & create interest to study that subject effectively



Significance of Results:

Assessment of effectiveness/Success of the activity:

With audio & video presentation the students got idea about different real time digital applications.

Reflective Critique:

Benefits:

Due to this activity the students are know about basic fundamentals of digital logic circuits & applications in digital world.

Challenges:

Need to pick best Video section with shorter delivery duration with clear audio.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO2	2	2	2	1		1			2	1		

CO1: The students will be able to apply the digital fundamentals in the present contemporary world.

Reference: <https://www.khanacademy.org/computing/computer-science/how-computers-work2/v/khan-academy-and-codeorg-circuits-logic>

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2019- 2020 (Odd Semester)

Innovative Practices Description

UNIT II COMBINATIONAL CIRCUIT DESIGN

Degree, Semester & Branch: III Semester B.E. ECE A

Course Code & Title: EC8392 Digital Electronics

Name of the Faculty member: Mrs.S.Jeeva

Name of the Topic: Adder & Subtractor

Name of the Innovative Practice: Simulation Tool

Date & Duration: 23.07.2019 & 10 minutes

ICT Tool Used: LCD Projector

Description:

ISim is a simulation tool integrated into Xilinx ISE. Like Modelsim and ActiveHDL simulation tool, ISim can be used by students to debug and verify their design. ISE Simulator is an application that integrates with Xilinx ISE to provide simulation and testing tools. Two kinds of simulation are used for testing a design: functional simulation and timing simulation. Functional simulation is used to make sure that the logic of a design is correct.

Goals (Learning Outcomes):

1. The students will be able to perform basic simulation tasks using ISim.
2. The students will be able to know the fundamentals of combinational circuits.

Use of Appropriate Method:

Justification for choosing the Simulation activity for this topic:

- It is a basic simulation tool, during higher semester the students are used this simulation tool at VLSI laboratory. So they must be able to know about it.

Effective Presentation:

Implementation (Plan & Execution) with Proof:

- First of all I explained about basics of Xilinx ISE simulator, types of coding & how to write the coding.
- Using that simulation software I typed the coding for basic combinational circuit of half adder by gate level modeling & behavioral level modeling and show the timing diagram.

- Similarly simulate the full adder, half subtractor, full subtractor & display the simulation results.



PROGRAM FOR HALF ADDER:

```
module halfadder (a,b,sum,carry);
    input a,b;
    output sum,
    carry;
    xor(sum,a,b);
    and(carry,a,b);
end module
```

```
module halfj(a, b, s, c);
    input a,b;
    output s,cout;
    assign sum=a^b;
    assign carry=a&b;
endmodule
```

PROGRAM FOR FULL ADDER:

```
module fulladder (a,b,cin,s,cout);
    input a,b,cin;
    output s,cout;
    wire p,q,r;
    xor1(p,a,b);
    xor2(s,p,cin);
    and1(q,a,b);
    and2(r,p,cin);
    or(cout,q,r);
end module
```

```
module fulladj(a,b,cin,s,cout);
    input a,b,cin;
    output s,cout;
    assign s=a^b^cin;
    assign c=(a&b)|(b&cin)|(cin&a);
end module
```

PROGRAM FOR HALF SUBTRACTOR:

```
module halfsubtractor (a,b,diff,borrow);  
    input a,b;  
    output diff,borrow;  
    xor(diff,a,b);  
    not(a1,a);  
    and(borrow,a1,b);  
end module
```

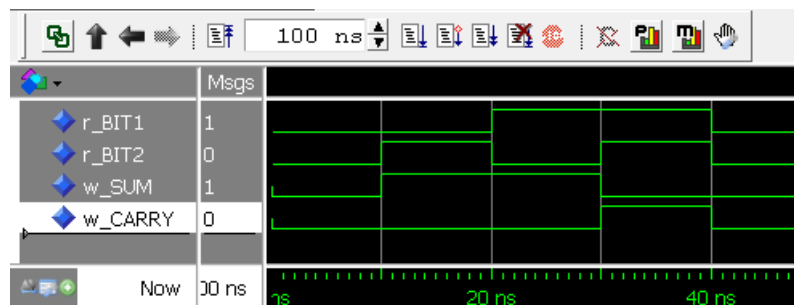
```
module halfj(a, b, diff, borrow);  
    input a,b;  
    output diff,borrow;  
    assign diff=a^b;  
    assign borrow=~a&b;  
endmodule
```

PROGRAM FOR FULL SUBTRACTOR:

```
module fullsubtractor (a,b,cin,diff,borrow);  
    input a,b,cin;  
    output diff,borrow;  
    wire p,q,r,s;  
    xor1(p,a,b);  
    xor2(diff,p,cin);  
    not1(q,a);  
    and1(r,q,b);  
    not2(s,p);  
    and2(w,s,cin);  
    or(borrow,w,r);  
end module
```

```
module fullsubtractor (a,b,cin,diff,borrow);  
    input a,b,cin;  
    output diff,borrow;  
    assign diff=a^b^cin;  
    assign borrow=(~a&b)||(~b&cin)||(~cin&a);  
end module
```

HALF ADDER SIMULATION:



Reflective Critique:

Benefits:

- Students are able to visually verify the sum & carry of adder, Borrow & difference of subtractor for all possible combinations in fraction of a second.

Challenges:

- Some of the students are struggle to differentiate various level of modeling techniques

Name of the Topic: Combinational Circuits

Name of the Innovative Practice: Online Quiz

Date & Duration: 05.08.2019 & 1 week

Reference: www.sanfoundry.com

Use of Appropriate Method:

Justification for choosing the Simulation activity for this topic:

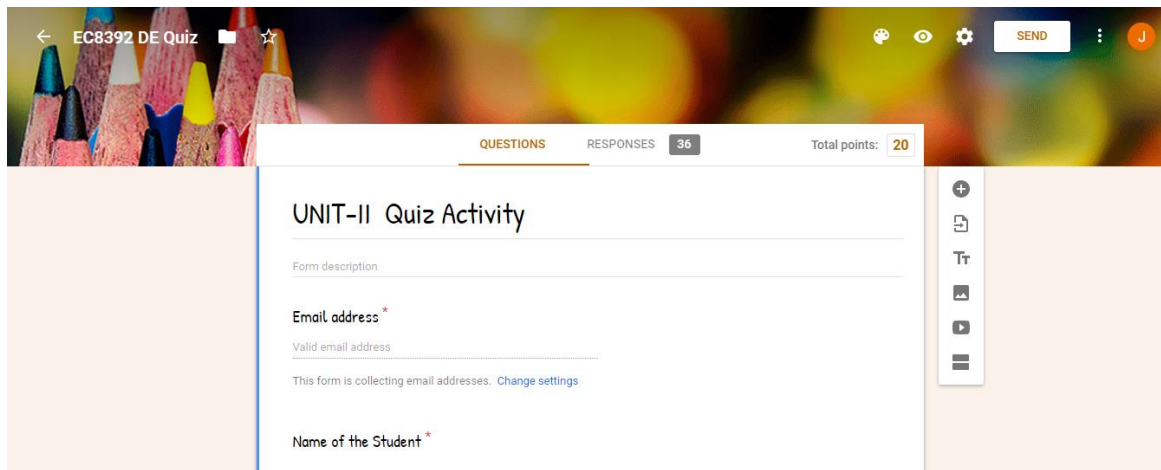
- This unit dealt with more number of combinational circuit examples. In higher degree entrance exam (TANCET, GATE) point of view students are must be known to all circuits. So I chosen this activity for revise the basic concepts.

Effective Presentation:

Implementation (Plan & Execution) with Proof:

- Using google form I sent questions (Multiple choice questions) to all students & also attached within mail. Each day will monitor the response of the students and release scores to their mail.

Effectiveness:



Reference: <https://www.sanfoundry.com/1000-digital-circuits-questions-answers/>

- 1) 3 bits full adder contains _____
 - a) 3 combinational inputs
 - b) 4 combinational inputs
 - c) 6 combinational inputs
 - d) 8 combinational inputs**Ans: d) 8 combinational inputs**

2) 2^9 input circuit will have total of _____

- a) 32 entries
- b) 128 entries
- c) 256 entries
- d) 512 entries

Ans: d) 512 entries

3) The number of logic gates and the way of their interconnections can be classified as _____

- a) Logical network
- b) System network
- c) Circuit network
- d) Gate network

Ans: a) Logical network

4) The simplified expression of full adder carry is _____

- a) $c = xy + xz + yz$
- b) $c = xy + xz$
- c) $c = xy + yz$
- d) $c = x + y + z$

Ans: a) $c = xy + xz + yz$

5) What is a multiplexer?

- a) It is a type of decoder which decodes several inputs and gives one output
- b) A multiplexer is a device which converts many signals into one
- c) It takes one input and results into many output
- d) It is a type of encoder which decodes several inputs and gives one output

Ans: b) A multiplexer is a device which converts many signals into one

6) In a multiplexer, the selection of a particular input line is controlled by _____

- a) Data controller
- b) Selected lines
- c) Logic gates
- d) Both data controller and selected lines

Ans: b) Selected lines

7) How many NOT gates are required for the construction of a 4-to-1 multiplexer?

- a) 3
- b) 4
- c) 2
- d) 5

Ans: c) 2

- 8) A combinational circuit is one in which the output depends on the _____
- a) Input combination at the time
 - b) Input combination and the previous output
 - c) Input combination at that time and the previous input combination
 - d) Present output and the previous output

Ans: a) Input combination at the time

- 9) The inputs/outputs of an analog multiplexer/demultiplexer are _____
- a) Bidirectional
 - b) Unidirectional
 - c) Even parity
 - d) Binary-coded decimal

Ans: a) Bidirectional

- 10) Most demultiplexers facilitate which type of conversion?
- a) Decimal-to-hexadecimal
 - b) Single input, multiple outputs
 - c) AC to DC
 - d) Odd parity to even parity

Ans: b) Single input, multiple outputs

- 11) In 1-to-4 demultiplexer, how many select lines are required?
- a) 2
 - b) 3
 - c) 4
 - d) 5

Ans: a) 2

- 12) How many AND gates are required for a 1-to-8 multiplexer?
- a) 2
 - b) 6
 - c) 8
 - d) 5

Ans: c) 8

- 13) Which IC is used for the implementation of 1-to-16 DEMUX?
- a) IC 74154
 - b) IC 74155
 - c) IC 74139
 - d) IC 74138

Ans: a) IC 74154

14) In 1-to-4 demultiplexer, if $C1 = 1$ & $C2 = 1$, then the output will be _____

- a) Y0
- b) Y1
- c) Y2
- d) Y3

Ans: d) Y3

15) How is an encoder different from a decoder?

- a) The output of an encoder is a binary code for 1-of-N input
- b) The output of a decoder is a binary code for 1-of-N input
- c) The output of an encoder is a binary code for N-of-1 output
- d) The output of a decoder is a binary code for N-of-1 output

Ans: a) The output of an encoder is a binary code for 1-of-N input

16) If we record any music in any recorder, such types of process is called _____

- a) Multiplexing
- b) Encoding
- c) Decoding
- d) Demultiplexing

Ans: b) Encoding

17) The primary use for Gray code is _____

- a) Coded representation of a shaft's mechanical position
- b) Turning on/off software switches
- c) To represent the correct ASCII code to indicate the angular position of a shaft on rotating machinery
- d) To convert the angular position of a shaft on rotating machinery into hexadecimal code

Ans: a) Coded representation of a shaft's mechanical position

18) One that is not the outcome of magnitude comparator is _____

- a) $a > b$
- b) $a - b$
- c) $a < b$
- d) $a = b$

Ans: b) $a - b$

19) Which one is a basic comparator?

- a) XOR
- b) XNOR

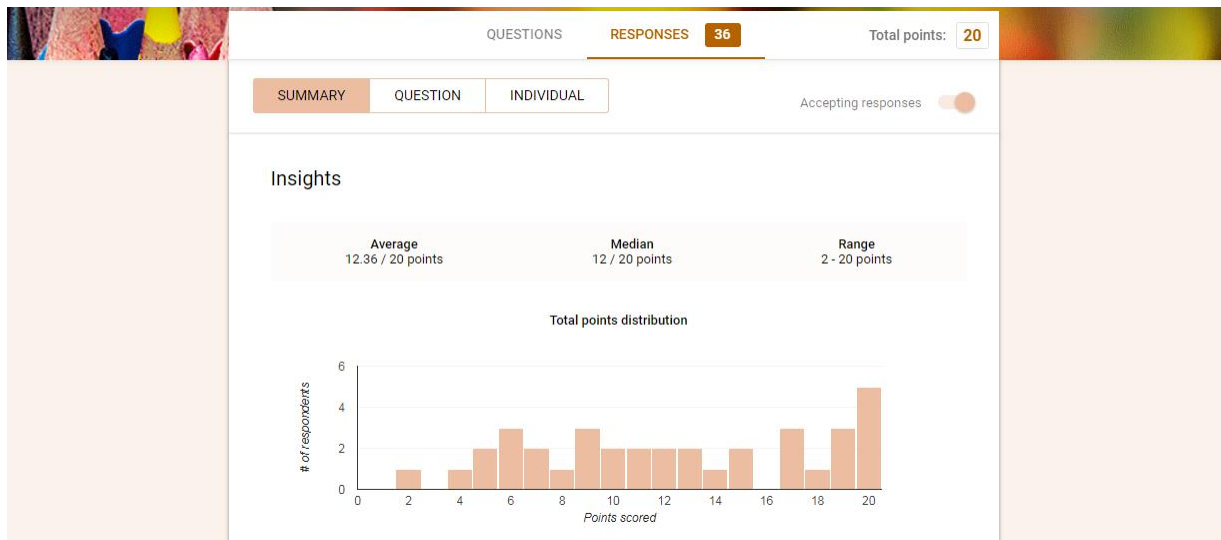
- c) AND
- d) NAND

Ans: a) XOR

20) Comparators are used in _____

- a) Memory
- b) CPU
- c) Motherboard
- d) Hard drive

Ans: b) CPU



Reflection Critique:

Benefits:

- ✓ Due to this activity students are recall the fundamentals of combinational circuits & it will be helpful for internal assessment test preparation.

Challenges:

- Within due to date they didn't attend so I extended that period.
- Some of the students are not interested to attempt the quiz, due to my compulsion they attended it. Even though one among them not yet to attend.

Significance of Results:

Assessment of effectiveness/Success of the activity:

Due to this activity, the students were able to recall the topics they have learnt in the classroom and they were also able to reproduce it in the exam. In the Internal Assessment Test-I most of the students attended the questions under this topic and scored good marks.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO2	2	2	2	1	3					1	1	1

CO2: The students will be able to design and implement various combinational digital circuits using logic gates.

References:

- 1) ISE Xilinx Simulator
- 2) <https://www.sanfoundry.com/1000-digital-circuits-questions-answers/>

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2019- 2020 (Odd Semester)

Innovative Practices Description

UNIT III SEQUENTIAL CIRCUITS

Degree, Semester& Branch: III Semester B.E. ECE A

Course Code & Title: EC8392 Digital Electronics

Name of the Faculty member: Mrs.S.Jeeva

Name of the Topic: Flipflops

Name of the Innovative Practice Roleplay

Date & Duration : 22.07.2019 & 15 Minutes

ICT Tools Used: White Board

Description:

An Approach to *Teaching and Learning. Role-play* is a technique that allows students to explore realistic situations by interacting with other people. It is a technique that allows students to explore realistic situations by interacting with other people in a managed way in order to develop experience and trial different strategies in a supported environment.

Goal (Learning Outcomes):

- ✓ The students are able to differentiate sequential & combinational circuits
- ✓ The students are able to acquire knowledge about flipflops

Use of Appropriate Method:

Justification for choosing the Role Play activity for this topic:

- Role play activity helps to develop their way of thinking, in this activity the students are think to design different sequential counters by using various flipflops.

Effective Presentation:

Implementation (Plan & Execution) with Proof:

- First I taught fundamentals & characteristics of sequential circuits
- Each students are assigned by one type of flip flop
- After that I called students one by one, they present concept of their own flip flop

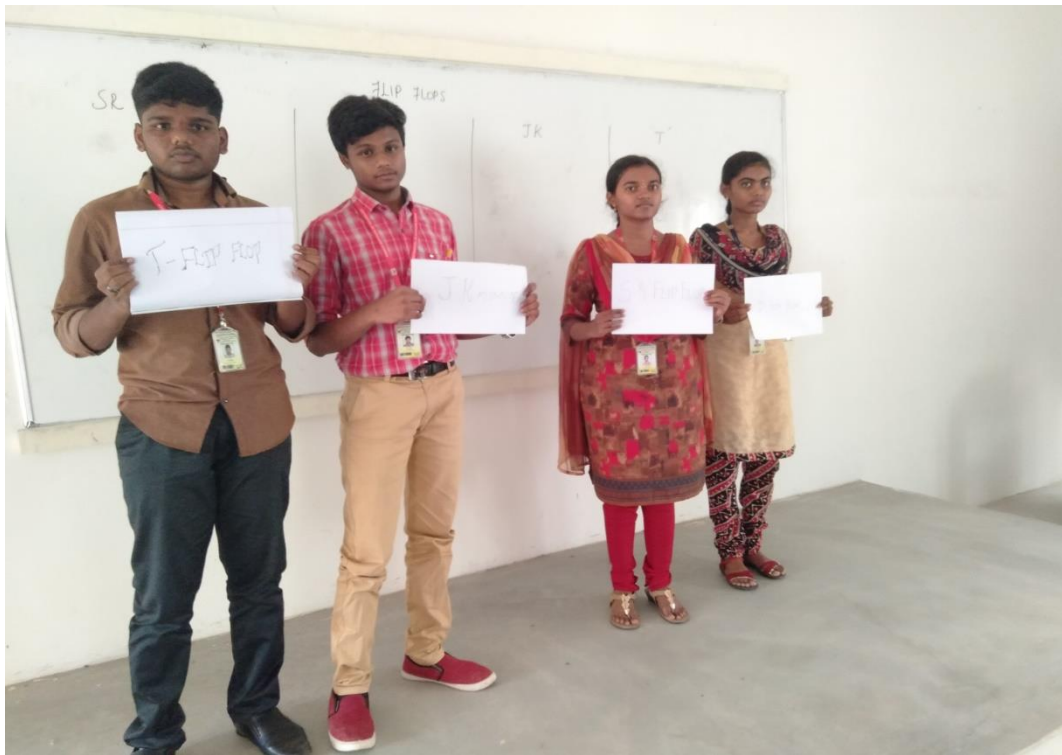


Fig 1: students are assigned by any one flip flop



Fig 2: Students present fundamentals of flip flops



Reflective Critique:

Benefits:

- Students are eagerly to present/act their role positively
- All the students are know the flip flop truth table, characteristic equation, state diagram & excitation table

Challenges:

- Some of the students are hesitate to present their role on dais. So, in future I will motivate those type students with giving special gifts like chocolates, pencil etc.

Name of the Innovative Practice: Mindmap

Name of the Topic: Sequential Circuits

Reference: M. Morris Mano and Michael D. Ciletti, “Digital Design”, 5th Edition, Pearson, 2014. (Page No.217)

Date & Duration: 22.07.2019 & 10 Minutes

Description:

Mind mapping is a visual form of note taking that offers an overview of a topic and its complex information, allowing students to comprehend, create new ideas and build connections.

Goal(Learning Outcomes):

- The students will be able to think different ideas to remember the concepts
- The students improve their self learning skills.

Use of Appropriate Method:

Justification for choosing the Mind map activity for this topic:

- Mind map is like a flow chart, so students are easily understand the design & analysis of sequential circuits
- This activity increases students creativity
- It is a simple way of revise the topics

Effective Presentation:

Implementation (Plan & Execution) with Proof:

- Two days before, I give the Mind map activity topic to the students.

- At the time of activity the volunteers they will come forward and present their flow chart to others.



Reflection Critique:

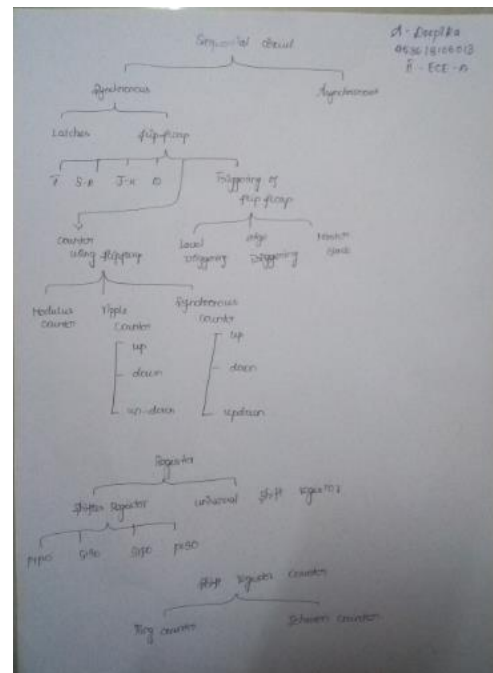
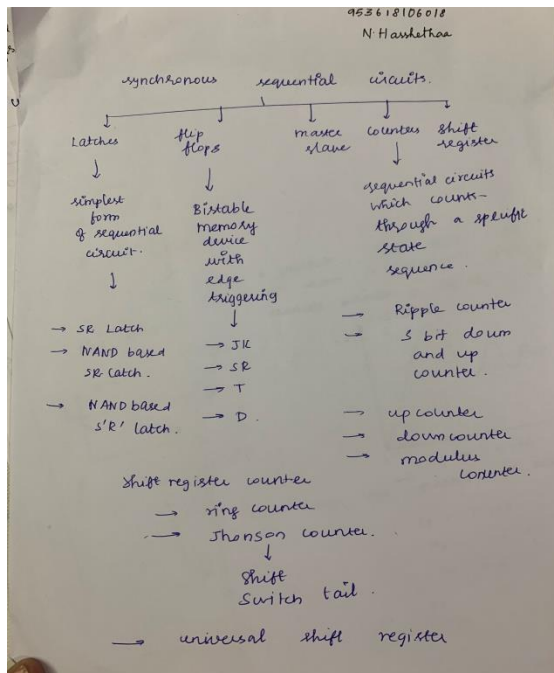
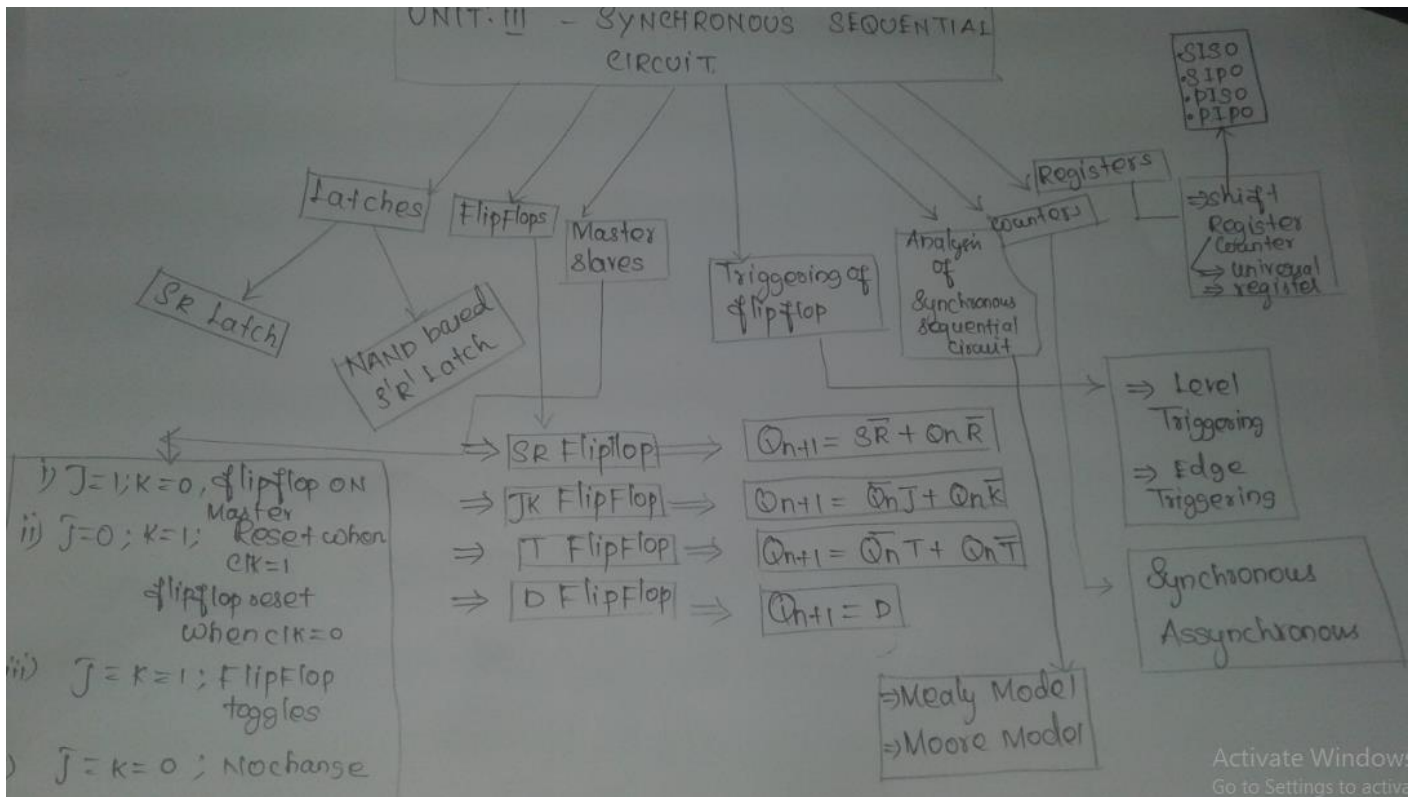
Benefits:

- ✓ Due to this activity the students are felt easy to understand the topics
- ✓ Students are actively participate and share their ideas in creative manner

Challenges:

- Some of the students are difficult to recollect all the points related to sequential circuits

Sample Mind Map Sheets:



Significance of Results:

Assessment of effectiveness/Success of the activity:

Due to this activity, the students were able to recall the topics they have learnt in the classroom and they were also able to reproduce it in the exam. In the Internal Assessment Test-II most of the students attended the questions under this topic and scored good marks.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO3	2	2	2	1	1					3		1

CO3: The students will be able to analyse and design synchronous sequential circuits

References:

- 1) S.Salivahanan and S.Arivazhagan,'Digital Circuits and Design' Fourth Edition, Vikas Publishing House Pvt. Ltd, 2012. (Page No.272).
- 2) M. Morris Mano and Michael D. Ciletti, "Digital Design", 5th Edition, Pearson, 2014. (Page No.217)

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2019- 2020 (Odd Semester)

Innovative Practices Description

UNIT IV ASYNCHRONOUS SEQUENTIAL CIRCUITS

Degree, Semester& Branch: III Semester B.E. ECE A

Course Code & Title: EC8392 Digital Electronics

Name of the Faculty member: Mrs.S.Jeeva

Name of the Topic: Introduction asynchronous sequential circuits

Name of the Innovative Practice: Think Pair Share

Date & Duration: 05.09.2019 & 15 Minutes

Description:

This technique gives students the opportunity to respond to questions in written form before engaging in meaningful conversation with other students. Asking students to write and discuss ideas with a partner before sharing with the larger group builds confidence, encourages greater participation, and results in more thoughtful discussions.

Think-Pair-Share (TPS) is a collaborative learning strategy where students collectively work together to solve a problem or answer to a reading assignment.

Goal (Learning Outcomes):

- The students will be able to differentiate the basic concepts about asynchronous sequential circuits from synchronous sequential circuits

Effective Presentation:

Implementation (Plan & Execution) with Proof:

1. **T :** (Think) First I'm asked question about fundamentals of asynchronous sequential circuits. Students "think" about what they know about the topic, because they have learned about synchronous sequential circuits.
2. **P :** (Pair) Each student should be paired with another student.
3. **S :** (Share) Students share their thinking with their partner. Finally I expand the "share" into a whole-class discussion.



Use of Appropriate Method:

Justification for choosing the Think Pair Share activity for this topic:

- This activity increases the involvement in listening & learning

Reflective Critique:

Benefits:

- ✓ Students are actively collaborate & share concepts with their pair students
- ✓ It increases the interaction with each other

Challenges:

- This activity conducted for whole class, I planned to conduct this activity in 15 minutes but consumes 20 minutes
- Most of the points are repeated

Name of the Innovative Practice: Random Name Picker

Date & Duration: 24.09.2019 & 10 Minutes

Name of the Topic: Problems in asynchronous sequential circuits (Cycles, Races, Hazards)

Reference: www.classtools.net

URL: https://www.classtools.net/random-name-picker/79_UbmEPN

Goal (Learning Outcomes):

- The students will be able to understand problems in asynchronous sequential circuits

Use of Appropriate Method:

Justification for choosing the Random Name Picker activity for this topic:

- The topic was very important in university exam so all the students are should be studied, so I chosen this game related activity to create interest on this topic.

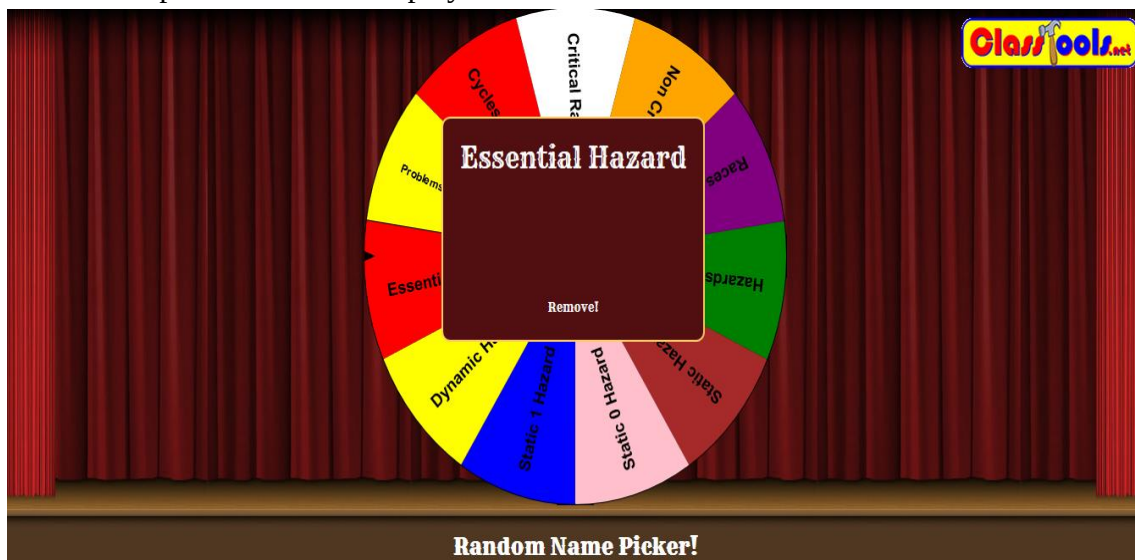
Effective Presentation:

Implementation (Plan & Execution) with Proof:

- Using Classtools.net, first I generate one random wheel for problems in asynchronous sequential circuits. It includes cycles, races, hazards. It is looks like a Spinning Wheel.



- I asked one student to come forward & click to spin the wheel. On Spinning the wheel the random problem will be displayed.



- After that the student discuss about that problem with other students. Once the problem selected that will be removed from the spinning wheel.



Reflective Critique:

Benefits:

- ✓ Due to this activity students are easily recall the concepts at the time of Internal Assessment Test III preparation.
- ✓ It will helps the students to improve their active learning.

Challenges:

- Only few number (10 students) of students got the chance to spin the wheel.

Significance of Results:

Assessment of effectiveness/Success of the activity:

Due to this activity the students are able to study the problems in asynchronous sequential circuits in easy manner.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO4	2	2	2	1	2				2	2		1

CO4: The students will be able to Design and implement asynchronous sequential circuits.

References:

- 1) M. Morris Mano and Michael D. Ciletti, "Digital Design", 5th Edition, Pearson, 2014. (Page No.474)

RAMCO INSTITUTE OF TECHNOLOGY
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Innovative Practices Description

Degree, Semester& Branch: III Semester B.E. ECE A

Course Code & Title: EC8392 Digital Electronics

Name of the Faculty member: Mrs.S.Jeeva

UNIT V MEMORY DEVICES AND DIGITAL INTEGRATED CIRCUITS

Activity: Crossword Puzzle

Date: 10.10.2019

Topic: Memory Devices

URL: <http://www.classtools.net/crossword/201910-dmHXVF>

Duration: 15 Minutes

Description: A crossword is an interesting word puzzle activity which helps the students to recollect the concept. Crossword puzzles provide students with an opportunity to evaluate their level of knowledge and to identify areas of further study in a fun way. Hence, including crosswords in the Engineering field teaching repertoire can be an innovative way to engage students in learning and help them to acquire knowledge and skills. It is one type of game, so this an enjoyable activity rather than regular multiple choice questions test.

Goal (Learning Outcomes):

- The students are able to remember the concepts at the time of examination
- The students are got confidence to study those topics

Effective Presentation:

Implementation (Plan & Execution) with Proof:

- ✓ First of all I am generate crossword puzzle by using classtool.net, it consists of 12 questions (Across-6 questions, Down-6 questions)
- ✓ The students are divided into groups (Boys-4 per team, Girls-6 per team)
- ✓ Individual Crossword puzzle sheets are provided to each group.
- ✓ The students answered the questions in the first 20 minutes of the class.
- ✓ The answer scripts are collected and evaluated by other group members.
- ✓ Finally I discussed the answers of the wrong questions at the end of the class.

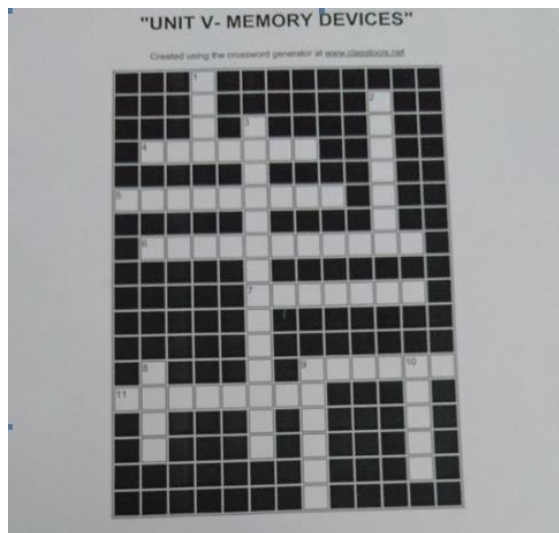


Fig 1: Crossword Puzzle Sheet

ACROSS

4. A place which is used as storage location in a computer _____ (7)

5. The DRAM stores its binary information on _____ (9)

6. CD-ROM is a _____ memory (11)

7. ROM is made up of OR gates & _____ (7)

9. MOS ROM is constructed using _____ (8)

11. Metal links are made up of _____ (5)

DOWN

1. A register is able to hold _____ (4)

2. In FPGA, vertical and horizontal directions are separated by _____ (7)

3. Main memories of a computer, usually made up of _____ (14)

8. Memories are classified into _____ categories (4)

9. The instruction used in a program for executing them is stored in the _____ (6)

10. How many $8\text{ k} \times 1$ RAMs are required to achieve a memory with a word capacity of 8 k and a word length of eight bits? (5)

For the interactive version, use this link: <http://www.classroom.net/crossword/201910-omHXVE>

Fig 2. Question Set



Fig 4: Students writing their answers

Question & Answers:

The instruction used in a program for executing them is stored in the

_____ ***MEMORY**

A register is able to hold _____ ***WORD**

When both the AND and OR are programmable, such PLDs are known as

_____ ***PLA**

Memories are classified into _____ categories ***FIVE**

Main memories of a computer, usually made up of _____ ***SEMICONDUCTORS**

ROM is made up of OR gates & _____ ***DECODER**

MOS ROM is constructed using _____ ***MOSFET**

The computer's main memory is ROM & _____ ***RAM**

How many $8\text{ k} \times 1$ RAMs are required to achieve a memory with a word capacity of 8 k and a word length of eight bits? ***EIGHT**

Metal links are made up of _____ ***NICHROME**

The DRAM stores its binary information on _____ ***CAPACITOR**

CD-ROM is a _____ memory ***NONVOLATILE**

A place which is used as storage location in a computer _____ ***ADDRESS**

The inputs in the PLD is given through _____ gates ***AND**

In FPGA, vertical and horizontal directions are separated by _____ ***CHANNEL**

Name of the Activity: Visual Quiz

Name of the Topic: Digital Integrated Circuits

Date: 18.10.2019

Description:

A Quiz is usually a form of a student assessment, but often has fewer questions of less difficulty and requires less time for completion than a test. A quiz is a form of game to test the students knowledge about a certain subject. A brief assessment used in education and similar fields to measure growth in knowledge, abilities and skills.

Effective Presentation:

Implementation (Plan & Execution) with Proof:

- First of all the students are divided into 8 small teams (Boys-3 teams, Girls-5 teams - 4 or 5 members per team)
- Option (A,B,C,D) Cutouts are given to each group
- Then I will display the Multiple choice question (MCQ) by using LCD projector at duration of 2 minutes.
- From each team any one of the student show the corresponding answer cutout.
- After that correct answers are discussed in class room itself.

THE OUTPUT OF THE 74 SERIES OF TTL GATES IS TAKEN FROM A BJT IN

- A. Totem pole and common collector configuration
- B. Either totempole or open collector configuration
- C. Common base configuration
- D. Common collector configuration

Fig 1. Multiple Choice Question displayed through LCD Projector

A CMOS IMPLEMENTATION OF A LOGIC GATE IS SHOWN IN THE FOLLOWING FIGURE: THE BOOLEAN LOGIC FUNCTION REALIZED BY THE CIRCUIT IS

Fig 2. Multiple Choice Question displayed through LCD Projector

Use of Appropriate method:

Justification for choosing Visual Quiz activity for this activity:

- To change the classroom teaching environment to different learning environment
- To create students collaboration within classroom



Fig 3. Students show the corresponding answer in the form of answer cutout



Fig 4. Students show the corresponding answer in the form of answer cutout

Reflective Critique:**Benefits:**

- ✓ Due to this activity the students are actively participated in group learning skill.
- ✓ Each students to improve their remembering capability.
- ✓ Easily to revise the fundamental concepts.

Challenges:

- Few students are not interested to hold the answer cut outs.

Significance of Results:**Assessment of effectiveness/Success of the activity:**

- Due to this activity, the students were able to recall the topics they have learnt in the classroom and they were also able to reproduce it in the exam. In the Internal Assessment Test-II most of the students attended the questions under this topic and scored good marks.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO5	2	2	2	1	1				2	2		1

CO5: The students will be able to apply the concepts of memory devices and programmable logic devices in Integrated Circuits

Reference:

- 1) S.Salivahanan and S.Arivazhagan,'Digital Circuits and Design' Fourth Edition, Vikas Publishing House Pvt. Ltd, 2012. & GATE Forum Materials